

A HUMAN-IN-THE-LOOP FRAMEWORK FOR CULTURALLY GROUNDED ELICITATION IN LINGUISTIC FIELDWORK IN INDIA

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Abstract

The elicitation of naturalistic linguistic data depends on the cultural and ecological suitability of the stimuli employed to prompt speaker responses. Standardized visual tools in language documentation are typically based on Western contexts, creating a mismatch in linguistically diverse settings such as India. This mismatch results in cognitive load and can affect elicited data due to increased code-mixing, reduced deictic accuracy, and interrupted speech. This paper puts forth a methodological proposal in the form of a Human-in-the-Loop (HITL) workflow for generating culturally grounded elicitation stimuli using generative AI. The approach formalizes iterative prompt engineering, ethnographic constraint specification, and pre-field validation to produce stimuli aligned with specific linguistic targets. Rather than treating generative models as autonomous systems, the framework positions the linguist as an active agent who constrains and evaluates outputs. The paper does not present an empirical evaluation; instead, it demonstrates the operational logic of the HITL methodology and documents the prompt design process in supplementary materials. The contribution is a replicable and transparent workflow for integrating generative AI into linguistic fieldwork.

Keywords: *Language Documentation, Generative AI, Human-in-the-Loop, Elicitation Methods, South Asian Languages*

1 Introduction

The elicitation of naturalistic linguistic data in fieldwork settings depends critically on the design of the stimuli used to prompt speaker responses. For decades, documentary linguistics has relied on standardized visual tools such as the *Pear Story* (Chafe 1980) and *Frog, Where Are You?* (Mayer 1969) to provide a controlled, non-linguistic baseline for cross-linguistic comparison. While methodologically convenient, these legacy stimuli are deeply embedded in Western ecological and cultural contexts, making them poorly suited for use in linguistically diverse settings such as India.

When deployed in Indian field contexts, such stimuli routinely introduce a mismatch between the visual environment and the lived experience of the speaker. This mismatch imposes additional cognitive load, shifting the task from spontaneous narrative production to the interpretation of unfamiliar objects, environments, and social relations (Abbi 2001; Lüpke 2009). The resulting data is systematically distorted: speakers exhibit increased code-mixing with dominant contact languages, reduced spatial and deictic precision, and higher rates of hesitation and syntactic fragmentation (Chelliah and de Reuse 2011; Bowern 2015). Consequently, the elicited corpus reflects artifacts of the elicitation method rather than the underlying grammatical competence of the speech community.

In the past, dealing with this issue in India meant making custom, culturally appropriate elicitation materials for different areas, ecologies, and speech communities. But these kinds of materials are expensive, take an extensive period of time, and are complicated to use across the subcontinent's many languages and cultures. There are other options, like participatory drawing and photography taken in that region, that are ecologically valid, but they come with logistical and ethical concerns that make it challenging to use them consistently in controlled elicitation tasks.

This paper proposes a methodological solution to this bottleneck: a formalized Human-in-the-Loop (HITL) workflow for generating culturally grounded elicitation materials using generative AI. The proposed approach does not seek to replace established fieldwork practices, but to complement them by providing a flexible, low-cost mechanism for rapidly generating context-specific stimuli where traditional resources are unavailable or impractical. The framework does not see generative models as independent tools. Instead, it sees the field linguist as an active editor who repeatedly limits, tests, and improves model outputs to meet specific linguistic and ethnographic needs.

The paper's scope is based on fieldwork in India, specifically the real-world problems that arose during data collection. The methodological principles delineated herein are designed to be applicable to other linguistically diverse contexts encountering analogous limitations. The paper does not provide an empirical assessment of this approach; rather, it illustrates the procedural rationale of the HITL methodology and showcases its application through organized, multimodal examples. Supplementary materials accompanying this paper provide full prompt transcripts, generation logs, and validation notes for all stimuli discussed.

2 The cultural bottleneck in elicitation design

2.1 Legacy stimuli and ecological mismatch

The primary function of visual stimuli in language documentation is to establish a controlled, non-linguistic baseline that enables comparable elicitation across speakers and languages (Lüpke 2009). For decades, this role has been fulfilled by standardized tools such as the Pear Story (Chafe 1980) and *Frog, Where Are You?* (Mayer 1969; Berman and Slobin 1994). These materials have been widely adopted because they provide a consistent narrative framework independent of any specific language.

However, these stimuli are grounded in mid-twentieth-century Western ecological and cultural settings. When used in Indian field contexts, they routinely introduce elements that are unfamiliar or irrelevant to the speaker's lived experience. Objects such as glass jars, bicycles, or specific agricultural practices may not map onto local material culture, while the social interactions depicted often diverge from locally meaningful norms. As a result, the stimulus ceases to function as a transparent elicitation device and instead becomes an object of interpretation (Bower 2015).

This mismatch shifts the task from narrative production to cognitive decoding. Rather than describing events in a familiar environment, speakers must first interpret an unfamiliar visual scene, which directly impacts the structure and content of the elicited data.

2.2 Intra-Indian variation and the problem of scale

The limitations of legacy stimuli are not resolved by replacing them with generic “Indian” alternatives. The Indian subcontinent is characterized by extensive variation in ecology, material culture, and social organization across language families and regions. A pan-Indian visual resource, often skewed toward a dominant North Indian, Hindi-speaking urban or semi-urban aesthetic, is fundamentally inadequate for rigorous elicitation. Indeed, generative models often prioritize these dominant cultural perspectives, struggling with the nuanced and contextual details of regional Indian traditions (Chhikara et al. 2025). A stimulus designed around a North Indian agrarian setting, for example, may be equally incongruent for speakers in coastal, forested, or mountainous environments.

Even within a single language family, micro-regional differences are significant. Variations in architecture, agricultural practices, clothing, and local ecology are tightly linked to lexical inventories and grammatical systems. A stimulus depicting wheat cultivation in North India will not provide appropriate lexical or semantic triggers for a speaker from a region where rice, millets, or plantation crops dominate.

This creates a fundamental scaling problem for elicitation design. Producing culturally appropriate stimuli for each speech community requires highly localized materials, but generating such materials through traditional means, manual illustration, staged photography, or participatory drawing is resource-intensive and difficult to standardize across field sites.

2.3 Linguistic consequences and limits of existing alternatives

The use of culturally incongruent stimuli is hypothesized to have direct and measurable consequences for the linguistic data elicited. When speakers are required to describe unfamiliar contexts, three recurring patterns emerge in the resulting discourse.

First, speakers exhibit increased reliance on contact-language forms, including loanwords and code-mixing, when local lexical equivalents are unavailable or difficult to retrieve (Abbi 2001). Second, when there aren't any culturally or spatially relevant reference points, spatial deixis and locative marking become less accurate, especially in languages that are known for making fine distinctions between different types of geography (Lüpke 2009). Culturally and topologically correct visual arrays are very important for getting accurate spatial information (Levinson 1992). Legacy stimuli often don't have these cultural affordances, so researchers have to hire local artists to change European-style images by hand, like getting rid of chimneys to make them more culturally relevant. This works, but it's challenging to achieve on a large scale (Webster et al. 2020). Third, the extra cognitive effort makes speech less fluent, with pauses, false starts, and simpler sentence structures, as speakers try to make sense of new visual information with their language system (Chelliah and de Reuse 2011).

Several alternatives have been proposed to address these limitations. Participatory approaches, in which community members produce drawings or photographs of local scenes, offer high ecological validity (Bowern 2015). But these methods also come with their own problems. The quality and clarity of drawings depend on the skill of the person making them. On the other hand, photographic methods need real people to work together and raise ethical concerns about privacy and representation. More importantly, neither method is flexible enough to consistently create controlled stimuli that focus on certain grammatical structures.

Because of this, there is still no way to make culturally relevant, linguistically focused elicitation materials that can be used in a wide range of field settings. It is this methodological gap that the present study seeks to address.

3 The Human-in-the-Loop (HITL) methodology

3.1 HITL as a methodological requirement

Generative AI models used for image and text generation operate through statistical associations between linguistic prompts and large-scale training data. They do not possess knowledge of linguistic structure, ethnographic context, or elicitation goals. When prompted in an unconstrained manner, such models default to dominant patterns in their training distribution, which are typically Western, urban, and standardized.

For language documentation, this presents a methodological risk. An unrestricted model shortfalls the ability to always generate stimuli that emphasize certain grammatical structures, encode contextually correct material culture, or represent sociolinguistic hierarchies related to a particular speech community. A HITL framework is therefore required to maintain cultural authenticity and prevent the misappropriation of linguistic data (Koc 2025). Within this framework, the linguist assumes the role of an active editor who constrains, evaluates, and iteratively refines model outputs. The objective is not visual realism per se, but the generation of stimuli that satisfy clearly defined linguistic and ethnographic requirements.

3.2 Iterative prompt engineering

The core mechanism of the HITL approach is iterative prompt engineering. Rather than relying on a single prompt, the linguist progressively refines input specifications to override the model's default assumptions and enforce culturally grounded constraints.

This process can be formalized as a sequence of stages:

a. Baseline prompt generation

A minimal prompt (e.g., “a woman cooking in a kitchen”) is used to establish the model's default output. This typically produces a generic or Westernized scene, revealing biases in the training data.

b. Macro-level specification

The prompt is elaborated to include broad regional identifiers (e.g., “an Indian woman cooking in a rural kitchen”). This results in some localization; however, the output still remains stereotyped or inconsistent.

c. Targeted ethnographic specification

The prompt is further refined to encode specific variables, including:

- material culture (e.g., tools, utensils, implements)
- spatial setting (e.g., courtyard, field, interior layout)

- bodily posture and action
- clothing and socially relevant markers

At this stage, the prompt is designed for elicitation relevance, ensuring that the generated scene can trigger the intended lexical or grammatical structures. Given the non-deterministic nature of generative models, the prompt transcript serves as the primary reproducible component of the methodology.

3.3 Controlling bias and structural hallucination

Even with detailed prompts, generative models introduce two persistent sources of error: representational bias and structural hallucination.

First, models tend to homogenize varied cultural contexts, generating stereotyped representations when prompted with broad terms like “rural India.” These outputs may combine incompatible elements from different regions or reflect media-influenced tropes rather than true circumstances. To address this, the HITL framework requires the use of explicit constraints and, where necessary, negative prompting to remove inappropriate features.

Second, models frequently generate structurally implausible objects or configurations. Tools may be rendered with impossible geometries, hybridized components, or incorrect functional properties. Such hallucinations are not just visual errors; they hinder elicitation by taking the speaker’s attention to the anomaly rather than the intended action or relation.

Within the HITL methodology, these risks are managed through systematic checks and revision of outputs. Any element that introduces ambiguity, implausibility, or cultural mismatch is dealt with prompt modification.

3.4 Pre-validation preparation: desk-based grounding

Prior to consultant validation, the HITL workflow incorporates a preparatory phase in which the linguist establishes a baseline understanding of the target community’s ecological, material, and sociocultural context. This stage draws on standard fieldwork practices, including literature review, prior field experience, and the consultation of available ethnographic or descriptive sources (Bowerman 2015; Chelliah and de Reuse 2011).

At this stage, the objective is not to generate stimuli but to define the constraints that will guide prompt engineering. Relevant information may include local agricultural practices, tools and technologies, domestic environments, clothing, and interactional norms. In addition, publicly available resources, including image databases and web-based materials, may be used to verify visual plausibility and refine prompt specifications.

Incorporating this preparatory knowledge into the prompt design process reduces dependency on trial and error during later stages. It enables the linguist to foresee likely sources of mismatch in model outputs and to encode more accurate constraints from the outset. This minimizes the number of revision cycles required during validation, hence improving both efficiency and consistency. This stage aligns the HITL workflow with established field preparation practices, ensuring continuity with available methodological standards.

3.5 Pre-fieldwork validation protocol

This phase is designed to assess the ethnographic adequacy of generated stimuli, rather than to collect linguistic data. This validation protocol reinforces the "epistemic responsibility" of the human expert to ensure representational accuracy before deploying synthetic stimuli in the field (Zhong et al. 2024).

A small number of native speakers are asked to evaluate the generated materials with respect to:

- accuracy of tools, objects, and environments
- plausibility of actions and postures
- appropriateness of social and spatial configurations

The task is explicitly framed as a critique rather than a description. Any stimulus that produces confusion, correction, or rejection is revised or discarded. Only stimuli that are accepted without hesitation proceed to elicitation contexts.

3.6 Offline deployment and field constraints

Generative AI systems typically require stable internet access and computational resources that are not available in many documentation settings, particularly in the Indian context. For this reason, stimulus generation, refinement, and validation are conducted before fieldwork. The finalized materials can then be organized into a static dataset, which can be deployed offline using printed images or preloaded digital devices.

This separation between generation and deployment keeps the elicitation process stable and repeatable while still taking advantage of the flexibility of AI-assisted stimulus design.

4 Technical demonstrations of the HITL workflow

This section demonstrates the operationalization of the HITL methodology through prompt refinement sequences. These examples are not presented as empirical evaluations but as controlled instances showing how iterative prompt refinement forces generic model outputs into linguistically targeted elicitation stimuli. These demonstrations include both successful refinements and cases where initial outputs required substantial revision.

4.1 Cooking scenario: eliciting event structure and verbal morphology

Domestic food preparation provides a productive domain for eliciting event structure, verb classes, and aspectual distinctions, particularly in relation to sequential and iterative actions. A baseline prompt, such as “a woman cooking in a kitchen” typically produces a generic, Western-style kitchen with modern cabinets, induction-compatible utensils, and electric stoves. Such scenes lack the material and spatial cues that can trigger the elicitation of culturally grounded lexical items and action descriptions.

Introducing a macro-level specification (e.g., “an Indian woman cooking in a rural kitchen”) results in partial localization but often produces stereotyped or internally inconsistent environments. These outputs may produce incompatible features, limiting their usefulness for elicitation.

Through targeted ethnographic specification, the prompt can be refined to encode:

- cooking infrastructure existing in that culture (e.g., clay or mud-based stove)
- spatial details (e.g., green courtyard or partially-open cooking area)
- Utensils used locally and fuel used for cooking
- bodily posture in manual food preparation (e.g., sitting on *modda*)

These constraints enable the formation of a familiar event frame in which participants, instruments, and sub-events are easily identifiable. As a result, the stimulus is better suited to eliciting predicates associated with preparation, transformation, and handling (e.g., cutting, kneading, roasting), as well as aspectual distinctions such as ongoing versus completed action (see Supplementary Material, Figure S1.1).

4.2 Agricultural activity: encoding argument structure and instrumentality

Agricultural scenes are frequently used to elicit verb morphology, argument structure, and case marking associated with manual actions and instruments. A baseline prompt such as “a farmer harvesting crops in a field” often produces mechanized or visually generic scenes, including tractors or non-specific tools, which obscure the intended agent–instrument–patient relations.

Even when broadly localized (e.g., “rural India”), such outputs do not reliably constrain the mode of harvesting or the tools involved and may include mechanized processes or structurally implausible implements.

Targeted prompt engineering allows the linguist to encode:

- crops grown in the area (e.g., wheat, paddy, millets)
- harvesting methods and technology commonly used
- the tools (e.g., curved iron sickle)
- posture of the body while cutting the crops
- appropriate time and weather

These details contribute to a clear agent–instrument–patient configuration, supporting the elicitation of transitive predicates, instrumental case marking, and verb agreement patterns. By constraining the visual input in this way, the stimulus aligns more closely with the semantic and syntactic structures under investigation. (See Supplementary Material, Case S1.2.1).

4.3 Social interaction and hierarchy: eliciting pragmatic and honorific systems

Eliciting expressions of social hierarchy, kinship, and honorific marking requires stimuli that encode culturally meaningful interpersonal relations. Generic prompts such as “a family talking at home” typically produce neutralized interactions lacking clear indicators of relative age, status, or social roles.

Even with broad contextual specification, such scenes often fail to represent socially structured asymmetries that are crucial for eliciting pragmatic distinctions.

Through iterative refinement, the linguist can introduce constraints that encode:

- generational asymmetry (e.g., elder versus younger participants)
- spatial positioning and body orientation (e.g., seated versus standing)

- culturally appropriate domestic settings
- interactional cues such as offering, requesting, or receiving objects

These features allow the stimulus to represent socially structured interactional contexts in which distinctions of respect, familiarity, and hierarchy are linguistically encoded. Such scenes are particularly suited to eliciting honorific morphology, pronominal variation, and pragmatic distinctions in address and reference.

By embedding these relational cues directly in the visual stimulus, the HITL workflow enables the elicitation of socially conditioned grammatical contrasts that are difficult to access through decontextualized prompts. (See Supplementary Material, Figure S1.2).

While the demonstrations above focus on visual stimuli, the HITL workflow is not limited to image generation. Text-based scenarios generated through LLMs can be combined with visual prompts to create multimodal elicitation contexts. In practice, visual stimuli can be employed to create a common situational framework, whereas text-based prompts may introduce regulated discourse conditions or narrative continuations. The combination of these modalities allows for more flexible targeting of grammatical structures across levels, including clause-level syntax, discourse sequencing, and pragmatic interpretation. (See Supplementary Material, Section 1.3).

5 Ethics and transparency in AI-assisted elicitation

5.1 Representational bias and cultural accuracy

Generative AI models learn from big datasets that show how visual and written content is spread out unevenly around the world. Because of this, illustrations of non-Western contexts are often not shown enough, look the same, or are seen through stereotypes that were made by people outside of the group. Models such as Stable Diffusion demonstrate a "rural-coded" bias towards Indian prompts, defaulting to impoverished or orientalist tropes even when urban contexts are specifically sought (Ghosh et al. 2024). When documenting languages, these biases can be methodologically and ethically risky, as they may produce inaccurate or externally imposed representations of the target community.

In the HITL framework, representational bias is seen as a limitation that should be actively managed rather than just a concomitant. The iterative prompt engineering process described in Section 3 functions, in part, as a mechanism for counteracting these biases by introducing explicit cultural and contextual constraints. However, the responsibility of correction remains with the linguist, who must evaluate generated outputs not only for visual plausibility but for ethnographic appropriateness.

This requirement highlights the importance of domain knowledge and pre-validation preparation. Without sufficient familiarity with the target community, the linguist risks reproducing the same distortions that the methodology seeks to avoid.

5.2 Ethical constraints in stimulus design

Beyond representational accuracy, the use of AI-generated stimuli raises broader ethical concerns related to the creation and use of visual materials in fieldwork.

Unlike photographic stimuli, which involve identifiable individuals and therefore raise issues of consent and privacy, AI-generated images allow for the creation of scenes without including real persons. This can reduce certain ethical risks, particularly in contexts where photographing participants is impractical or frowned upon. However, it also brings new concerns.

First, the generation of culturally specific scenes without direct community involvement raises questions about authorship and representation. The HITL methodology somewhat addresses this by having a validation phase in which native speakers validate the generated materials. Nevertheless, the linguist's interpretation and available knowledge still drive the initial construction of stimuli.

Second, there is a risk of over-standardization. In attempting to produce “representative” scenes, the linguist may mistakenly suppress intra-community variation, presenting a single version of cultural practice as normative. This is particularly relevant in linguistically and culturally diverse settings, where variation is itself a central subject of inquiry.

The HITL workflow takes generated stimuli as elicitation tools rather than documentary representations to deal with these risks. Its goal is to prompt linguistic structures, not to supply ethnographic descriptions. This difference limits the interpretive weight placed on the images and reduces the risk of misrepresentation.

5.3 Transparency, reproducibility, and citation practices

A central methodological challenge in using generative AI is the non-deterministic nature of model outputs. Similar prompts may give different results across sessions, and model behavior may change over time as systems are updated. This raises concerns regarding reproducibility, which is a core requirement of scientific methodology.

Within the HITL framework, reproducibility is addressed through the systematic documentation of prompts and generation conditions, which are stored in the Supplementary Material. This record allows other researchers to reconstruct the decision-making process underlying stimulus design, even if exact visual replication is not possible.

In addition, model details, generation dates, and relevant constraints are documented to the extent possible. While generative systems do not always provide stable versioning comparable to traditional software, recording available metadata boosts transparency and allows for partial replication of results. This approach shifts the locus of reproducibility from the generated visuals themselves to the prompting and refinement process, treating prompt design as the central unit of the framework.

A further requirement of transparency concerns the citation of generative AI systems used in the production of elicitation materials. Unlike conventional tools, generative models contribute directly to the creation of research stimuli, making their role methodologically significant. However, citation practices in linguistics have not yet standardized how such tools should be documented.

Under HITL, generative systems are considered methodological tools, and their use should be reported. At a minimum, this entails model name, access platform, date of generation, and prompt (see Supplementary Material). Standardized citation of AI-assisted stimulus generation promotes transparent elicitation conditions.

6 Conclusion

This paper proposes a Human-in-the-Loop framework for generating culturally relevant elicitation stimuli with generative AI, taking into account the constraints of linguistic fieldwork in India. The basic assumption is that the effectiveness of an elicitation task is not only dependent on the quality of the stimuli but also on how well it matches the social, material, and environmental setting of the speech community. Legacy stimuli, which are standardized and convenient, have been ineffective in diverse linguistic settings. The HITL framework addresses this problem through a formal framework in which the linguist iteratively engineers the stimuli, grounds it ethnographically, and also pre-validates it. Instead of using generative AI as a standalone technology, the framework seeks to integrate it with traditional linguistic fieldwork practices. The demonstrations presented in this paper have illustrated how such an approach can be used to construct elicitation stimuli that are better aligned with specific linguistic targets, including event structure, argument structure, and socially conditioned grammatical distinctions.

The contribution of this study is methodological rather than empirical. It does not describe the effectiveness of the AI-produced stimuli quantitatively. It provides a workflow for the application, testing, and future improvement of the application. It also provides a method for keeping the generative systems open and reproducible through concise documentation, validation, and citation.

In the future, the application could be improved through a systematic pilot study design comparing the effectiveness of AI-assisted stimuli with traditional stimuli. Shared resources of culturally relevant stimuli and images for less-documented languages could be created. This would expand the evaluation possibilities while still allowing for the flexibility necessary to adapt the application to different languages and cultures.

Data availability statement

The complete Prompt Transcript Appendix and the original high-resolution visual stimuli generated for this study are openly available alongside this preprint as supplementary files on Zenodo at: <https://doi.org/10.5281/zenodo.19161560>

Declaration of generative AI in scientific writing

The core arguments, methodological frameworks, and theoretical insights presented in this paper are the original work of the authors, derived directly from qualitative observations and linguistic fieldwork. During the drafting of this manuscript, generative AI (OpenAI's ChatGPT-4 and Google Gemini) was utilized strictly for language polishing, syntactic refinement, and structural formatting to meet academic standards. The authors manually reviewed, edited, and verified all text, and assume full and sole responsibility for the integrity and accuracy of the published content.

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